

Orthogonal Trust: The Independence of Privacy and Accuracy in Instructor Evaluations of AI Tools

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Abstract

Educational institutions face a tension between generative AI's affordances and the compliance risks of sending student data to third-party cloud APIs. On-device small language models (SLMs) offer an alternative, guaranteeing data sovereignty through local execution, yet pedagogical viability remains unestablished. We present TOOL, an on-device educational assistant that runs a 1.5-billion-parameter model locally without cloud connectivity, with no-code fine-tuning on instructor materials and built-in Socratic guardrails. Automated benchmarks show the educator-evaluated model achieved 12.4 times higher guidance ratings than its base model while preserving aggregate MMLU performance. To evaluate deployment readiness, we conducted a qualitative case study with seven educators spanning computing, engineering, and education-policy contexts who integrated their own course materials and evaluated TOOL through think-aloud interaction, interviews, and a 30-item survey. Our findings reveal orthogonal trust: data privacy and epistemic accuracy function as independent, non-fungible evaluation axes. Educators decisively endorsed the local privacy architecture ($M = 4.07$), but this did not transfer to confidence in factual reliability ($M = 2.74$); parametric leakage and fluent-but-wrong output led six of seven educators to reject unverified adoption, and five of seven identified traceable citations as non-negotiable for classroom use. While privacy is necessary for institutional consideration, epistemic reliability remains the binding constraint on adoption.

Notes for Practice

- **Context & Empirical Problem:** Institutional policies often treat privacy and pedagogical reliability as substitutable. This study shows they function as *orthogonal trust dimensions*: educators strongly endorse on-device data sovereignty, but this does not transfer to factual reliability, which remains the binding barrier to adoption.
- **For System Designers:** Resolving data privacy does not inherently earn instructional trust. Interfaces for on-device small language models should prioritize verifiable citation traceability and explicit uncertainty signaling to reduce cognitive audit friction.
- **For Educational Institutions & Procurement:** Local-first edge deployment resolves third-party data transmission risks but is not a comprehensive institutional safeguard. Privacy architecture and epistemic verification should be evaluated as independent, non-fungible criteria.
- **For Pedagogical Deployment:** Because parametric leakage and hallucination hinder both instructional preparation and direct student inquiry, current-generation edge SLMs require structured human-in-the-loop verification rather than assuming readiness for unmonitored classroom use.
- **For Learning Analytics Researchers:** A bounded case-study design that triangulates interaction data, fieldnotes, and survey items – and treats low construct-level internal consistency as an analytic finding rather than noise – can surface decoupled trust dimensions that an aggregate score would miss.

Keywords

trust, privacy, accuracy, artificial intelligence in education, on-device language models, educator evaluation, learning analytics

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1. Introduction

Large language models (LLMs) are transforming the landscape of automated instructional support in higher education. In computing education specifically, emerging generative architectures provide continuous debugging assistance while suppressing direct code revelation (Liffiton et al., 2023), scaffold problem-solving through iterative code reconstruction (Kazemitabaar et al., 2025), and align interactive guidance with formal pedagogical taxonomies (Maurya et al., 2025). As the field increasingly recognizes, classroom AI adoption is fundamentally socio-technical: instructional efficacy depends not on raw foundation model benchmarks alone, but on contextualized interface design, verifiable student data governance, and instructor agency in steering algorithmic behavior (Nazaretsky, Ariely, et al., 2022; Yan et al., 2024).

However, mainstream educational AI tooling remains almost universally dependent on centralized, commercial cloud APIs. This reliance introduces two systemic challenges that threaten institutional adoption. First, client-server architectures necessitate transmitting student prompts, telemetry, and proprietary coursework to external commercial servers. In the learning analytics (LA) community, such third-party data pipelines raise immediate alarms regarding student surveillance, corporate data harvesting, and compliance with statutory mandates such as FERPA and GDPR (Furze & Roe, 2024; Slade & Prinsloo, 2013). Regulatory friction has already prompted administrative enforcement against at least one commercial LLMs over data protection violations (Kuru, 2024), while institutions face mounting ethical obligations to safeguard learner privacy and maintain algorithmic sovereignty. Second, commercial frontier models are optimized via reinforcement learning from human feedback (RLHF) for immediate perceived helpfulness, functioning effectively as automated answer keys. This default behavior short-circuits the productive struggle and metacognitive monitoring essential for novice programmers (Pitts et al., 2025; Prather et al., 2024). Under automation bias (Buçinca et al., 2021), learners tend to prompt repeatedly for direct solutions rather than engaging with reflective scaffolding (S. Yang et al., 2024).

Small language models (SLMs) running locally on edge hardware offer an architectural alternative. By executing inference entirely on-device, compact models (1B–7B parameters) ensure that learner interactions and course materials never leave local memory, bypassing third-party cloud infrastructure entirely (Clear, 2025; Dikici & Bilgin, 2025; Wang et al., 2025). Yet deploying SLMs in educational settings presents a sharp technical tension: compact models lack the broad parametric reasoning of frontier cloud models and exhibit elevated hallucination rates on complex, multi-step tasks (Srivastava et al., 2025). Furthermore, customizing local models typically demands advanced machine learning expertise in parameter-efficient fine-tuning (PEFT), presenting a substantial usability barrier for non-technical educators (Celik, 2023; Hedderich et al., 2024).

To examine this socio-technical trade-off, we developed TOOL, an on-device, privacy-preserving educational assistant. TOOL combines a 1.5-billion parameter base model (Qwen2.5-1.5B-Instruct) executed via Apple MLX with a no-code drag-and-drop workflow that allows instructors to upload course PDFs and trigger automated, local Low-Rank Adaptation (LoRA) fine-tuning directly on their laptops. The fine-tuned adapter is engineered with built-in Socratic guardrails that withhold direct answers, redirect off-topic queries, and generate stepwise hints grounded in uploaded curriculum materials. We deployed TOOL as a technology probe in an in-depth qualitative case study with seven computing educators ($N = 7$) spanning undergraduate, graduate, and professional instructional roles. Our inquiry is guided by five research questions:

- **RQ1 (Trust Calibration):** What factors govern computing educators’ trust in an on-device, document-grounded AI assistant, and what evidence standards must be satisfied before permitting student deployment?
- **RQ2 (Accuracy & Leakage):** How faithfully does the local SLM adhere to instructor-provided curriculum materials, and how do domain experts detect and evaluate parametric confabulations?
- **RQ3 (Pedagogical Fit):** To what extent does the on-device Socratic facilitation model support authentic instructional goals without collapsing into answer-giving?
- **RQ4 (Privacy & Sovereignty):** How does local-first edge execution influence educator comfort with data exposure relative to centralized cloud alternatives?
- **RQ5 (Adoption Readiness):** What socio-technical and operational barriers determine whether educators would integrate local SLM tooling into their authentic workflows?

Our findings reveal the empirical reality of orthogonal trust: data privacy and epistemic accuracy operate as decoupled, non-fungible evaluation axes. While educators decisively endorsed the local-first privacy architecture ($M = 4.07/5.00$), this assurance did not compensate for the model’s factual unreliability and susceptibility to parametric leakage ($M = 2.74/5.00$; 100% observed hallucination rate). Solving data sovereignty is an essential prerequisite that grants local AI tools institutional consideration, but it remains insufficient for adoption. Educators demand verifiable accuracy, identifying traceable source citations as the non-negotiable bridge between edge privacy and instructional trust.

2. Related Literature

2.1 From Automated Answers to Scaffolded Guidance

Generative models are now capable enough to answer most introductory programming exercises outright (Finnie-Ansley et al., 2022), and unconstrained deployments of this capability have been shown to induce metacognitive deskilling and to widen, rather than close, learning disparities (Bastani et al., 2025; Prather et al., 2024). Evaluating frontier models against a pedagogical taxonomy, Maurya et al. (2025) found that systems routinely act as answer keys rather than tutors. In response, scaffolding-oriented systems such as CodeHelp (Liffiton et al., 2023) and CodeAid (Kazemitabaar et al., 2024) constrain generation toward hints rather than solutions, and Socratic-dialogue framings (Sunil & Thakkar, 2025) attempt to preserve productive struggle by design. Even so, human oversight of these systems remains fallible: Ahmed et al. (2025) found that Socratic prompting suppressed solution leakage but that human teaching assistants still failed to reliably detect hallucinated content. For learning analytics, the implication is that scaffolding design choices are not sufficient on their own to guarantee trustworthy analytics-informed instruction; the analytics and evaluation layer that surfaces model errors to stakeholders matters as much as the scaffolding logic itself.

2.2 Data Sovereignty, Privacy, and Governance

Most generative learning applications rely on cloud-hosted foundation-model APIs, transmitting student telemetry, prompts, and coursework to third-party commercial servers. Within learning analytics, this has raised sustained concern over surveillance, data extraction, and compliance with statutory mandates such as FERPA and GDPR (Garcia et al., 2025; Slade & Prinsloo, 2013; Yan et al., 2024), and regulatory friction has already produced at least one administrative action against a commercial LLM over data-protection violations (Kuru, 2024). Institutional data-governance frameworks are increasingly formalizing resistance to sharing sensitive learner data with third parties (Furze & Roe, 2024; Shahzad et al., 2025), and cloud-only architectures structurally exclude bandwidth-limited or air-gapped learning contexts (Mimoudi, 2025). Community-based models for expanding technology access have similarly called for AI tooling that is low-barrier and deployable without dependence on institutional cloud infrastructure or specialized technical staff (Johnson et al., 2024), motivating architectures that decouple instructional AI assistance from centralized cloud dependencies entirely.

2.3 Teacher-AI Teaming and the Dimensions of Trust

Instructor trust is a stronger predictor of classroom AI adoption than demographic factors or raw model capability (Nazaretsky, Ariely, et al., 2022; Nazaretsky et al., 2025; Viberg et al., 2025). Celik (2023) and Celik et al. (2025) identify trust as driven primarily by *pedagogical reliability* (valid, non-hallucinatory guidance) and *privacy protection* (safeguards against data surveillance), and this dual structure is echoed in the broader learning-analytics literature on educator adoption, where perceived usefulness and perceived risk load onto separate factors (Viberg et al., 2025). Related work has examined this same tension from the student side, asking whether a generative AI tool functions as a genuine instructional resource or a superficial novelty (Cora et al., 2025); we instead probe this question from the instructor's vantage point, where misplaced trust puts an entire classroom at stake rather than one learner. At the same time, this literature has generally examined these dimensions in aggregate, or has studied them as they arise in cloud-hosted systems where privacy and accuracy risks are bundled into the same third-party vendor relationship. Whether the two dimensions remain independent once privacy risk is architecturally removed altogether, as with a fully on-device system, to our knowledge has not been directly tested. We term this decoupled relationship **orthogonal trust**: the socio-technical condition in which an instructor's confidence in a system's data privacy operates independently from their confidence in its inferential accuracy and pedagogical correctness. Understanding how educators negotiate this divergence, rather than assuming privacy and accuracy trade off or reinforce one another, is central to designing learning-analytics tools that educators and institutions can adopt with informed confidence, and speaks directly to this venue's interest in the institutional and ethical conditions under which analytics-driven tools can be trusted at scale.

2.4 On-Device Small Language Models

Small language models (SLMs, roughly 1–7B parameters) offer one technical path toward resolving the privacy tension above by executing inference entirely on local hardware rather than a vendor's cloud (Clear, 2025; Dikici & Bilgin, 2025; Wang et al., 2025). Parameter-efficient fine-tuning and distillation have narrowed the capability gap between SLMs and frontier cloud models on domain-restricted tasks (Srivastava et al., 2025), and a small number of studies have piloted local, offline deployments in educational settings (Reza et al., 2025; Yu et al., 2025). These studies have focused on latency, throughput, and automated rubric scoring; how educators themselves judge deployment readiness once local fine-tuning happens on their own device – and whether removing cloud transmission changes their calculus around accuracy risk – remains an open, and squarely learning-analytics-relevant, empirical question.

The Present Study. TOOL addresses this gap directly: an on-device workflow in which instructors upload their own curriculum materials to trigger local adaptation of a small base model under strict pedagogical guardrails, evaluated not on

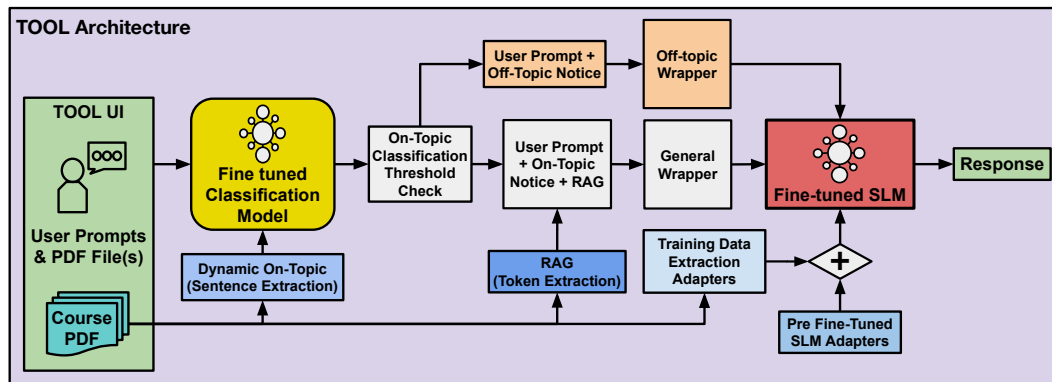


Figure 1. TOOL system design and architecture pipeline for active on-device small language fine-tuning.

throughput but on whether computing educators across a range of instructional roles find the resulting system trustworthy enough to consider adopting.

3. System Design & Technical Architecture

3.1 Design Goals

TOOL was developed to explore whether a locally deployed small language model (SLM) can provide pedagogically useful support in education without cloud infrastructure or technical expertise from the instructor (Bastani et al., 2025; Liffiton et al., 2023). TOOL is a research prototype built to investigate our research questions around trust, not a production system; we prioritize investigating the viability of the concept over production-level engineering, and leave performance optimization, scalability, and broader model evaluation to future work. The design is organized around three goals drawn from the literature: (i) preserve learning by withholding direct answers (Bastani et al., 2025; Liffiton et al., 2023), (ii) encourage productive struggle through step-wise scaffolding rather than complete solutions (Kazemitabaar et al., 2024), and (iii) eliminate privacy risk by keeping all computation and student data on the instructor's device (Kasneci et al., 2023). Our focus here is on whether the combination of on-device fine-tuning, pedagogical guardrails, and no-code instructor customization produces a system that educators find trustworthy enough to consider deploying.

3.2 System Overview

TOOL is a modular, locally deployable pipeline that runs entirely on the instructor's own machine, with no third-party cloud dependence and no remote API calls. Illustrated in Figure 1, the system has four functional layers. A lightweight chat interface lets instructors and students converse with the assistant; it performs no inference itself, only presenting the conversation and routing messages to a local backend process. That backend orchestrates a small (1.5-billion-parameter) base language model, which is fetched once and cached locally rather than bundled, so the system footprint stays small. When an instructor uploads course materials (e.g., PDFs), the text is chunked and embedded in memory at runtime, giving the assistant a retrieval-augmented, document-grounded context window without requiring a persisted vector database. Incoming prompts pass through a lightweight on-topic classifier before generation: prompts judged relevant to the uploaded materials are answered with retrieved context injected; off-topic prompts trigger a trained refusal-and-redirect behavior instead of a generic answer.

During development we also produced a second, more heavily constrained model variant for a specific programming course; it appears only in the automated benchmarks (Section 4) for comparison and was not used in the expert evaluation.

3.3 Model Customization Without Code

Both model variants are built on the same openly available 1.5-billion-parameter base model, chosen because models at this scale retain useful procedural and logical reasoning within a restricted domain while still fitting on consumer laptop hardware (Srivastava et al., 2025; A. Yang et al., 2024). Rather than requiring instructors to write code or manage a training pipeline, TOOL performs parameter-efficient (LoRA) fine-tuning locally and automatically once an instructor uploads a small set of curated example interactions for their course; a system prompt persona enforces that the model teaches only within the instructor's stated domain, provides scaffolded rather than complete solutions, and declines off-topic requests. This lets an instructor specialize the assistant to their own curriculum without surrendering pedagogical control to a general-purpose cloud model, and without needing machine-learning expertise themselves.

3.4 Feasibility on Consumer Hardware

Because the system is intended for instructors' own laptops rather than institutional servers, we benchmarked both fine-tuning and inference on ordinary consumer hardware (an 8 GB-memory laptop, no discrete GPU). Fine-tuning a course-specific adapter

completed in under an hour of active computation, and inference ran at an interactive pace (roughly 25–28 tokens/second) with a peak memory footprint of about 3.4 GB — within reach of typical instructor hardware, though with the kind of run-to-run timing variance one would expect from a memory-constrained consumer device rather than dedicated training infrastructure. We report the full training corpus composition, hyperparameters, per-category inference benchmarks, and hardware logs in the Supplementary Materials at the end of this paper, since these implementation specifics matter for replication but are secondary to the trust-evaluation questions this paper investigates.

4. Model Benchmark Evaluation

4.1 Technical Evaluation

4.1.1 Knowledge Preservation (MMLU)

To understand how fine tuning affects TOOL’s response quality and reasoning, we use the MMLU (Massive Multitask Language Understanding) benchmark to determine how well the models perform on an array of different subjects ranging from STEM to humanities and difficulty levels ranging from elementary to advanced professional level (Hendrycks et al., 2021). We found that both TOOL and its base model, Qwen2.5-1.5B-Instruct, performed very similarly, achieving scores of 55.7 and 55.9 respectively. Both scores reflect evaluation within our local test harness (step-by-step system prompt, ANSWER-format extraction). Because this harness yields a base score below the officially reported 60.9, we interpret only the within-harness comparison, where the 0.2-point difference falls within expected sampling variation.

4.1.2 Instructional Guidance Quality (LLM-as-a-Judge)

We are ultimately interested in whether our TOOL provides better *guidance*, teaching correct concepts through sound reasoning without directly providing complete answers (Maurya et al., 2025). Since the MMLU system prompt instructs models to “think step by step before answering” and provide a final answer in the format “ANSWER: [LETTER]”, we evaluated the quality of the step-by-step reasoning preceding the answer.

We employed an LLM-as-a-judge approach using two evaluator models with different rubric stringency:

Llama 3.1 70B Instruct (Content-Focused Rubric): Content-focused evaluation emphasized the factual value of responses:

- **Yes (Score 3):** Accurate explanations, sound reasoning, teaches correct concepts
- **To some extent (Score 2):** Has explanations but contains errors or misconceptions
- **No (Score 1):** Just states answer without explanation

Claude Sonnet 4.6 (Pedagogy-Focused Rubric): We also applied a stricter pedagogy-focused rubric. This evaluator considered both the quality of the reasoning and whether the response explicitly disclosed the selected answer before the required final-answer line:

- **Yes (Score 3):** Accurate explanations, sound reasoning, provides correct concepts, doesn’t provide complete answer
- **To some extent (Score 2):** Has explanations but contains errors, misconceptions, or complete answers
- **No (Score 1):** No meaningful guidance (e.g., only a letter answer)

The pedagogy-focused rubric therefore treats answer disclosure as a limitation even when the surrounding explanation is conceptually correct. An answer presented only in the required final-answer line is not penalized.

Human Validation of Automated Guidance Ratings. Four members of the author team reviewed usable guidance labels to 1,140 of the 1,199 sampled responses (95.1%) using the same written three-level rubric. The initial content-focused evaluation achieved 75.1% exact agreement with the human ratings ($n = 1,135$; Cohen’s $\kappa = 0.544$); five responses were excluded because API overload errors prevented the automated evaluator from returning valid labels. The stricter Claude Sonnet 4.6 pedagogy-focused evaluation achieved 61.0% exact agreement ($n = 1,138$; Cohen’s $\kappa = 0.317$) after excluding one safety refusal and one internally inconsistent judgment. Because the stricter evaluator generally assigned lower ratings than the human reviewers, we treat it as a conservative sensitivity analysis. Additional agreement statistics are reported in the Supplementary Materials at the end of this paper.

Table 1. Benchmark results comparing instructional guidance quality across models. Ratings are generated using the Llama 3.1 70B content-focused judge.

Model	Answer Correctness	Yes	To Some Extent	No
Constrained course-specific variant (§3.2; benchmarks only)	Correct	19.8%	63.3%	17.0%
	Incorrect	1.0%	69.6%	29.4%
Qwen2.5-1.5B-Instruct	Correct	2.5%	40.1%	56.9%
	Incorrect	0.3%	41.0%	58.2%
TOOL (evaluated in §§5–6)	Correct	33.1%	53.6%	12.8%
	Incorrect	1.4%	64.0%	33.3%

4.1.3 Benchmark Results

The MMLU results indicate that fine-tuning preserved the base model’s general knowledge performance. Across the complete automated benchmark, TOOL received a *Yes* rating of 12.4 times the base model rate under the stricter guidance rubric.

Across the automated guidance evaluations, TOOL received stronger guidance ratings than the base model (Table 1). We also examined guidance ratings conditional on final-answer correctness to distinguish useful reasoning from merely selecting the correct option. Percentages are calculated within each model-by-answer-correctness stratum using all evaluated responses as the denominator. Unresolved Claude judge errors, representing up to 1.3% of the strata shown, are not displayed; consequently, some rows sum to less than 100%.

The resulting pattern suggests that the fine-tuning increased the prevalence of useful intermediate reasoning, including among some responses with incorrect final answers. We interpret this as evidence about the instructional characteristics of the generated responses, not as evidence of improved student learning. The human-validation results further indicate that this conclusion is strongest under the content-focused rubric and more conservative under the stricter pedagogy-focused rubric.

5. Human-Centered Qualitative Case Study

To evaluate how educators assess, calibrate, and act upon automated, on-device pedagogical analytics, we conducted a qualitative case study (Creswell & Poth, 2016; Yin, 2018) grounded in human-centered learning analytics (HCLA) methodologies. Rather than treating AI adoption as an abstract usability metric, our evaluation centers on the educator as a primary stakeholder in data-informed instructional workflows, examining whether TOOL meets the threshold conditions of epistemic trust and data protection required before closing the feedback loop to learners. Each educator constituted a bounded case, enabling rich, contextualized cross-case comparison across our five research questions.

Seven computing educators (P1–P7) were recruited via purposive sampling (Patton, 2014). Participants were selected as domain experts in their respective computing instructional areas who self-identified as non-experts in machine learning architecture or model development. This profile reflects the target stakeholder population who would deploy and audit local AI analytics directly from course syllabi and slides without technical machine learning assistance. Participants represented diverse instructional contexts across higher education, professional training, and community education (see Table 2).

Table 2. Participant demographics and background ($N = 7$). Details anonymized.

ID	Instructional Context	Domain	Document(s) Used
P1	Graduate computing course	Machine learning	Research paper; textbook chapter
P2	Professional development	Structural engineering (code-based)	Engineering code book
P3	Community-based technical education	Introductory programming	Lecture materials; study guide
P4	K–12 educational support	Educational policy / technology	Policy document
P5	Classroom / instructional	Data structures & algorithms (CS2+)	Course PDF; pasted content
P6	Personal / individual use	Data science / machine learning	Bullet-point document
P7	Graduate computing course	Research methodology	Research paper

Domain reflects the subject-matter area of the participant’s uploaded course materials, not the participant’s personal AI/ML expertise.

Procedure & Think-Aloud Interaction. Each participant completed a ~60-minute remote session conducted by two researchers and recorded with consent. Following a document-grounding tour, participants interacted with TOOL using their own course materials while thinking aloud (Ericsson & Simon, 1993), completed a semi-structured interview (Kvale & Brinkmann, 2009), and completed the post-session instrument. The Session Lead facilitated the interaction, while an Observer recorded timestamped field notes concerning verification behavior, hesitation, repair strategies, and changes in expressed trust (Denzin, 2017; Lincoln, 1985). Complete timings, prompts, and observation protocols are provided in the Supplementary Materials.

Multi-Stream Data Collection & Instruments. We triangulated eight distinct data streams: verbatim session transcripts, independent observer fieldnotes, raw system chat logs, screen recordings, pre-session baseline surveys, post-session evaluative surveys, researcher reflexive memos, and member-checking summaries.

The evaluation employed a 30-item Likert instrument (5-point scale: 1 = “Strongly Disagree” to 5 = “Strongly Agree”) distributed across six a priori constructs (Nazaretsky, Cukurova, & Alexandron, 2022): *Accuracy and Faithfulness*, *Trust and Confidence*, *Perceived Usefulness*, *Response Quality*, *Pedagogical Appropriateness*, and *Privacy and Data Comfort* (see Table 3). The post-session instrument paired these rating scales with structured open-ended prompts capturing qualitative narratives, failure workarounds, and data boundary reflections.¹ Full survey instruments and interview question banks are provided in Appendix A and Appendix B.

Table 3. Construct-level descriptive statistics ($N = 7$). k = number of items; α = Cronbach’s alpha.

Construct	Items	k	M	Mdn	SD	α
Accuracy & Faithfulness	Accurate responses, Correct attribution, Preserved nuances, Code syntactically correct, Accurate CS explanations, Avoided hallucination	6	2.74	2	0.73	.84
Trust & Reliability	Trust outputs, Confidence matched quality, Uncertainty visible, Trust without verification, Handled edge cases	5	2.77	3	0.83	.79
Perceived Usefulness	Useful in real work, Saved time, New perspectives on docs, Useful for materials prep, Useful for student Q&A, Would recommend	6	2.71	3	0.64	.67
Pedagogical Suitability	Appropriate for students, Respected pedagogical intent, Supports scaffolding, Trust for direct student use	4	3.04	3	0.51	.52
Response Quality	Clear & well-organized, Appropriate vocabulary, Appropriate detail level, Source info helpful, Appropriate context amount	5	3.26	3	0.59	.49
Privacy & Data Comfort	Comfortable uploading docs, Local-first increases willingness, Compatible w/ institution policy, Comfortable w/ student info	4	4.07	4	0.53	.47

M and SD were computed directly from the raw Likert export (Exit Survey responses), using participant P6’s first submission where a duplicate submission occurred (see Methods).

Stepwise Qualitative & Computational Analysis. To reconcile inductive concept discovery with our deductive research questions, we employed an integrated stepwise analytic strategy combining automated text mining, multi-coder thematic analysis, and joint display integration:

- **Exploratory Concept Mapping (Leximancer 5.0):** We imported a consolidated corpus of interview transcripts, observer notes, chat logs, and open-ended survey responses into Leximancer 5.0 (Lemon & Hayes, 2020; Sotiriadou et al., 2014). Interviewer dialogue tags and protocol noise terms were excluded. Using a hybrid User Defined + Automatic configuration, the algorithm computed unsupervised word co-occurrences around 50 researcher-defined seed concepts derived from our RQs and LA literature, generating an objective, auditable spatial concept map that sensitized subsequent manual coding.
- **Iterative Qualitative Coding (MAXQDA):** Qualitative data streams were organized into seven participant document groups within MAXQDA. Three researchers independently coded transcripts and observer notes using a deductive codebook anchored to RQs 1–5 alongside an inductive open-coding scheme, meeting iteratively to reconcile discrepancies. Observer fieldnotes received parallel behavioral codes (e.g., [ANALYTIC], trust shifts, verification strategies) to track gaps between reported attitudes and observed actions.
- **Cross-Case Synthesis & Joint Displays:** We utilized MAXQDA’s Code Matrix and Code Relations Browsers to generate case-by-theme matrices. Quantitative survey scores were linked to each case via document variables, producing joint displays that paired composite Likert scores directly with contextualized quotes on identical dimensions (Creswell & Clark, 2018).

6. Findings: Privacy and Accuracy as Orthogonal Trust Dimensions

This section presents mixed-methods findings from our qualitative evaluation study ($N = 7$). We triangulate our 30-item Likert survey data with open-ended qualitative reflections, observer fieldnotes, chat logs, and systematic cross-case thematic matrices. The analysis is structured around five core evaluation dimensions: trust calibration (RQ1), accuracy and parametric leakage (RQ2), pedagogical fit (RQ3), privacy and data comfort (RQ4), and adoption readiness (RQ5). Throughout, we integrate sensitivity analysis to identify items where ordinal (median) and interval (mean) interpretations diverge, highlighting individual risk tolerances within this bounded sample.

¹To maintain temporal consistency, all analyses utilize the initial post-session survey completed immediately upon session conclusion; one participant asynchronously submitted an unprompted re-assessment four days later, which was excluded from our statistics to preserve standardized measurement timing.

6.1 Construct-Level Overview

Table 3 summarizes the six Likert constructs, revealing two overarching structural patterns. First, *Privacy and Data Comfort* emerged as the highest-rated construct (composite $M = 4.07$, $Mdn = 4.00$), demonstrating decisive educator endorsement of TOOL's local-first edge execution. Second, *Accuracy and Faithfulness* ($M = 2.74$, $Mdn = 2.00$) and *Perceived Usefulness* ($M = 2.71$, $Mdn = 3.00$) were the lowest-rated dimensions, reflecting persistent educator apprehension regarding the SLM's factual reliability and hallucination frequency. Between these poles, *Trust and Confidence* ($M = 2.77$, $Mdn = 3.00$), *Pedagogical Appropriateness* ($M = 3.04$, $Mdn = 3.00$), and *Response Quality* ($M = 3.26$, $Mdn = 3.00$) clustered near the neutral scale midpoint. Internal consistency was strong for *Accuracy and Faithfulness* ($\alpha = 0.84$) and *Trust and Confidence* ($\alpha = 0.79$), indicating that participants evaluated these core dimensions with high conceptual coherence. Lower alpha values for the remaining dimensions (0.47–0.67) reflect heterogeneous, context-dependent priorities across teaching roles. With $N = 7$, alpha point estimates carry wide uncertainty and are reported descriptively rather than as formal scale reliability metrics, as visualized in the construct distribution analysis (Figure 2).

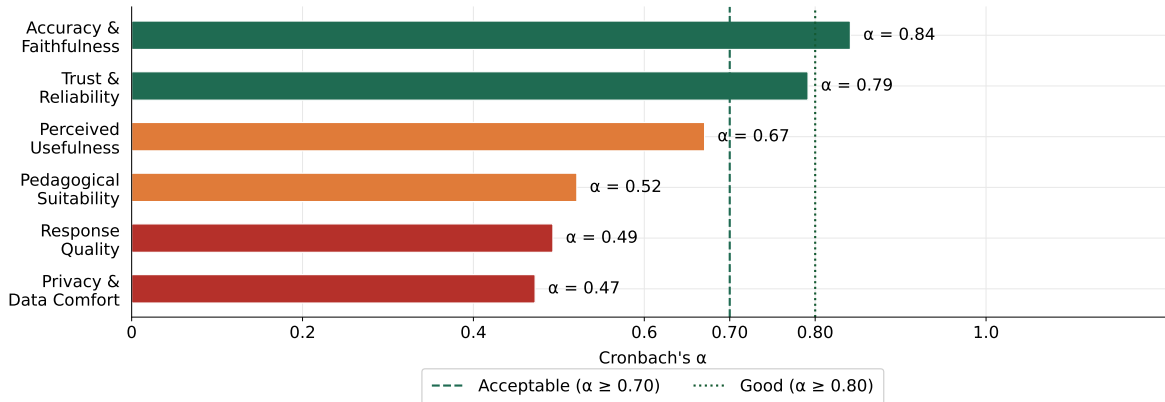


Figure 2. Internal consistency (Cronbach's α) across Likert constructs ($N = 7$). Accuracy and Faithfulness ($\alpha = 0.84$) and Trust and Confidence ($\alpha = 0.79$) exceeded the $\alpha > 0.70$ reliability threshold.

6.2 Trust Is Conditional and Evidence-Dependent (RQ1)

Across sessions, participants characterized trust as an active, conditional evaluation rather than a categorical default. Four of seven participants (P1, P3, P4, P7) systematically cross-checked model generations against their original PDFs or domain expertise, treating TOOL as a fallible starting point for inquiry. For instance, P2 probed whether TOOL could locate textual evidence within course source code:

“I wanted to see if the tool would be able to read the code and point me to specific sections relevant to questions I asked. While the tool would point me towards a section of the code, it was often incorrect or misinterpreted.” —P2

Similarly, P1 tested conceptual consistency by requesting technical illustrations of concept drift:

“I asked the assistant questions to clarify my understanding of the document... Initially, the responses were reasonable and helpful, but when I asked the assistant for examples on a concept called concept drift, it seemed to provide examples that did not match my understanding based on the textbook.” —P1

Survey metrics corroborate this skepticism. The item assessing unverified trust (Q10: “*I would trust the assistant enough to use its output without additional verification*”) yielded the second-lowest score across the instrument ($M = 2.00$, $Mdn = 2.00$); only Q6, avoiding ungrounded information, scored lower ($M = 1.86$), with six of seven participants selecting Disagree or Strongly Disagree. Conversely, conversational tone (Q8, $M = 3.29$) and edge-case handling (Q11, $M = 2.71$) received moderate ratings, underscoring that conversational polish failed to compensate for epistemic uncertainty.

6.2.1 Traceable Citations as a Non-Negotiable Prerequisite

Five of seven educators explicitly identified grounded, traceable citations as the definitive capability required to make the tool viable for instructional deployment. Participants positioned source attribution as an essential mechanism for verifying correctness:

“I need citations throughout answers to cross-check.” —P7

For other educators, citation traceability was tied directly to professional accountability:

“I want to see where information was pulled from. I need 100% accuracy.” —P4

“I always appreciate active citations and sources; it reduces my hesitation about hallucination.” —P3

Rather than viewing factual errors as fatal dealbreakers, educators framed trust as an auditable verification workflow. P5 articulated a quantitative risk threshold, stating the tool would be viable if its hallucination rate remained reliably below 20%. However, because TOOL currently lacks automated passage-level attribution, the cognitive burden of auditing outputs rested entirely on the instructor.

6.3 Document Grounding Does Not Prevent Parametric Leakage (RQ2)

Accuracy and Faithfulness emerged as both the most internally consistent ($\alpha = 0.84$) and most heavily criticized dimension ($M = 2.74$). Item Q6 (“*The assistant avoided generating information not present in my documents*”) was the most consensual item in the survey: All participants (7/7) participants rated it negatively ($M = 1.86$, $SD = 0.38$). In every session, the SLM introduced facts, assumptions, or code patterns originating from its pre-training corpus rather than the uploaded PDF.

Participants documented recurring failure modes across disciplines. P7 observed severe scope creep where the model expanded beyond the source boundary:

“It would bring in outside information in addition to what I asked. Sometimes it would answer the question on the first bullet point then give additional information that was not accurate.” —P7

P3 noted ungrounded introductory programming examples, while P5 observed document retrieval breakdowns where the model answered only when excerpts were pasted directly into the chat prompt. These cross-domain breakdowns confirm that parametric leakage is an architectural limitation of compact parameter weights rather than a domain-specific quirk.

Crucially, participants highlighted a disconnect between factual precision and stylistic presentation. While *Response Quality* rated moderately high ($M = 3.26$, driven by Q18 on organization [$M = 3.71$] and Q19 on technical vocabulary [$M = 3.57$]), this linguistic fluency actively disguised factual distortions. As P2 summarized:

“The explanation was fine, just wrong.” —P2

This stylistic masking introduces acute pedagogical risks: learners lacking advanced domain expertise are vulnerable to accepting syntactically polished hallucinations.

Notably, educators evaluated code generation (Q4, $M = 3.29$, zero negative ratings: 6/7 Neutral, 1/7 Strongly Agree) far more favorably than natural language synthesis (Q1–3, Q5–6, all below scale midpoints). Observer fieldnotes indicate this gap was driven by verification friction: while natural language confabulations were immediately visible against an instructor’s domain knowledge, syntactically clean code required active sandbox execution to reveal runtime errors or subtle edge-case omissions.

6.4 Socratic Design Avoids Answer-Giving but Lacks Guided Depth (RQ3)

Pedagogical Appropriateness clustered around the scale midpoint ($M = 3.04$) and demonstrated low internal consistency ($\alpha = 0.52$). All participants rated respect for pedagogical intent exactly Neutral (Q24, $M = 3.00$, $SD = 0.00$), while ratings of scaffolding support were also centered at Neutral (Q25, $M = 3.00$, $SD = 0.58$). The fine-tuned model consistently avoided direct answer-dumping, but it struggled to sustain dynamic, multi-turn Socratic inquiry.

Educator willingness to deploy the tool in student-facing environments was highly polarized. Some participants supported bounded access restricted to low-stakes clarification:

“I would share it with students for simple, basic questions at their level of questioning.” —P3

“I would share it directly with students [for FAQ-style policy questions].” —P4

Conversely, other educators warned that novice learners lack the metacognitive grounding to detect subtle distortions:

“Students without baseline knowledge would be confused.” —P1

This polarization was evident on Q16 (*usefulness for answering student questions*; $M = 2.71$, $Mdn = 2.00$): four participants rated it negatively while three rated it positively, with zero neutral selections. Sensitivity analysis confirmed student deployment decisions depend on individual instructor risk tolerance regarding hallucination rather than broad pedagogical disagreement.

6.5 Educators Embrace Local-First Privacy but Guard Student Data (RQ4)

Privacy and Data Comfort was the highest-rated construct ($M = 4.07$, $Mdn = 4.00$) and the sole dimension achieving decisive consensus. All participants (7/7) agreed or strongly agreed that they felt comfortable uploading course documents to the local model (Q27, $M = 4.29$, $SD = 0.49$; 5/7 Agree, 2/7 Strongly Agree). Knowledge of on-device processing actively increased adoption willingness for six of seven participants (Q28, $M = 4.43$, $Mdn = 5.00$; 4/7 Strongly Agree). Furthermore, six of seven educators agreed that local-first execution aligned with institutional AI governance (Q29, $M = 4.14$).

Qualitative feedback emphasized data sovereignty and independence from commercial vendor tracking:

“I like not giving info to big tech.” —P2

“It make[s] me feel like I can trust the tool more knowing it would not be released to the ‘cloud.’” —P4

However, the low internal consistency of this construct ($\alpha = 0.47$) exposed a critical privacy boundary: comfort dropped substantially when shifting from instructor-authored teaching materials to student data. Item Q30 (*comfort uploading de-identified student work*; $M = 3.43$, $SD = 1.27$) revealed strong divergence. As P1 reflected:

“Knowing my data only stays on my device would make me more willing to use the tool with materials that may contain more sensitive information. That being said, I would still be pretty hesitant to use it since I am not sure how the information is being used and stored.” —P1

Thus, while local execution eliminates third-party transmission risks, it does not alleviate broader institutional anxieties surrounding on-device caching, local persistence, and data governance.

6.6 Adoption Readiness and Emergent Workaround Strategies (RQ5)

Despite strong enthusiasm for local privacy, TOOL did not cross the threshold for regular workflow integration. No participants (0/7) agreed that the assistant currently saves time (Q13, $M = 2.43$; 1/7 Strongly Disagree, 2/7 Disagree, 4/7 Neutral), and willingness to recommend the tool skewed negative (Q17, $M = 2.43$; 2/7 Strongly Disagree, 1/7 Disagree, 3/7 Neutral, and 1/7 Agree). The cognitive overhead of verifying every output neutralized the efficiency gains of automated guidance.

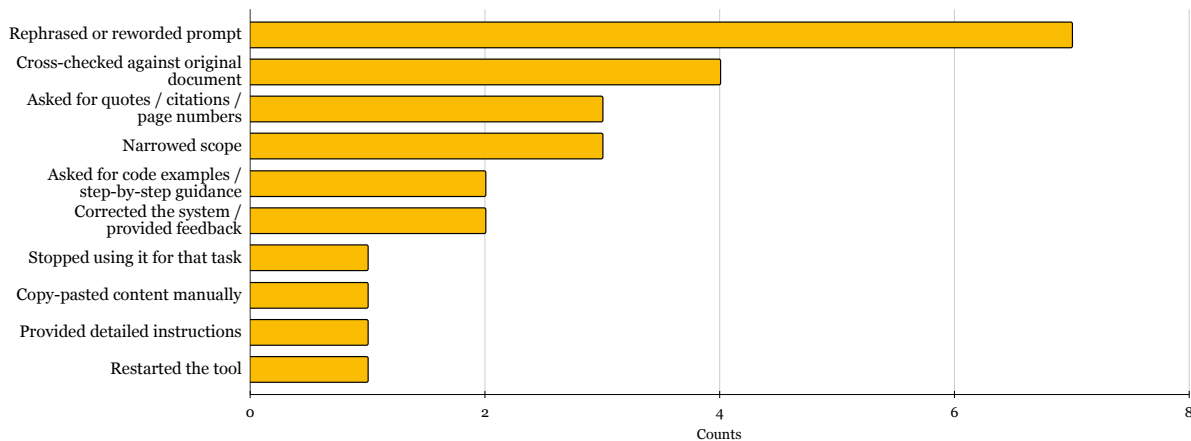


Figure 3. Observed workaround strategies across sessions ($N = 7$). Rephrasing queries was universal (7/7), while manual cross-checking was deployed by four of seven participants (4/7) and prompt narrowing by three (3/7).

When encountering hallucinations, participants developed structured workaround strategies (Figure 3). Query rephrasing occurred universally across all seven sessions (7/7), with three participants (P1, P3, P7) independently converged on *prompt narrowing*, or the decomposing of broad questions into granular queries targeting specific subheadings or localized paragraphs:

“When I became more specific in my questions, asking about specific subheadings rather than the overall document, I received more responses.” —P3

This behavior shows users adapting naturally to SLM parameter ceilings by narrowing the context window’s search space.

6.7 Cross-Case Thematic Synthesis

A cross-case thematic matrix synthesizing our qualitative coding and Leximancer relational analysis surfaced four core socio-technical themes:

1. **The Orthogonality of Privacy and Accuracy:** Resolving institutional data privacy through edge computing is an enabling condition that grants TOOL a hearing, but it does not advance epistemic trust. Educators treat data protection and pedagogical correctness as decoupled, non-fungible criteria.
2. **Superficial Fluency and Automation Risk:** Professional typography, coherent markdown structuring, and academic vocabulary mask underlying parametric leakage. This stylistic polish increases cognitive audit friction for educators and poses significant failure modes for novice learners.
3. **Citations as the Bridge to Usability:** Traceable, passage-level citations were identified by five of seven participants as the single missing capability needed to make an on-device SLM deployable, converting invisible hallucination audits into tractable fact-checking.
4. **Adaptive User Constrained Prompting:** In response to bounded 1.5B model capacity, educators organically develop prompt-narrowing strategies, compensating for limited inferential reasoning by providing tighter syntactic and semantic constraints.

These themes capture the central paradox of current edge AI in education: the local architecture that successfully solves institutional privacy compliance simultaneously limits the parameter capacity needed to guarantee factual fidelity.

7. Discussion

This qualitative evaluation demonstrates that educator trust in on-device educational AI is bifurcated across two independent evaluation axes. While participants decisively endorsed the local-first architecture for safeguarding data sovereignty, they rejected the model's epistemic reliability due to severe parametric leakage and hallucination rates. These findings confirm that data privacy and factual accuracy operate as orthogonal trust dimensions: resolving institutional compliance and student surveillance concerns does not compensate for, or advance, confidence in model correctness.

Our findings extend educational AI adoption models (Celik et al., 2025; Nazaretsky, Cukurova, & Alexandron, 2022; Viberg et al., 2025) by demonstrating that even when privacy concerns are resolved via local execution, epistemic failures independently block classroom integration. Although the Socratic design successfully prevented direct solution leakage, participants encountered severe automation bias risks (Buçinca et al., 2021): professional markdown formatting, polished syntax, and academic vocabulary masked underlying confabulations. Consistent with findings on teaching assistants evaluating LLM hints (Ahmed et al., 2025), instructors least familiar with the underlying course materials were the most vulnerable to these fluent errors. Furthermore, participants adapted to the bounded reasoning of the 1.5B model by organically converging on prompt-narrowing strategies, mirroring user adaptations observed in cloud-hosted systems such as CodeHelp (Liffiton et al., 2023) and CodeAid (Kazemitabaar et al., 2024).

These socio-technical tensions suggest four design imperatives for educational SLMs: (1) **First-class citation infrastructure:** Automated passage-level attribution is essential to convert opaque verification into manageable fact-checking. (2) **Task-constrained scoping:** Edge SLMs should target bounded, syntactically verifiable tasks rather than open-ended dialogue, supported by rubrics to detect downstream code vulnerabilities. (3) **Explicit uncertainty signaling:** Visual confidence cues are necessary to prevent uniform conversational fluency from disguising confabulations (Ujváry et al., 2026). (4) **Conservative deployment boundaries:** Because both instructor preparation and student-facing interactions face high error rates, uncritical assumptions that classroom AI adoption is inherently desirable must be tempered.

Institutions could scale local deployment through shared curricular adapters that instructors customize locally. Ultimately, on-device SLMs solve the institutional data-sovereignty bottleneck, but local execution cannot substitute for epistemic validity; accuracy remains the binding constraint on pedagogical adoption. We frame the four points above as design considerations surfaced by this sample, not as generalizable prescriptions; a bounded qualitative case study of seven educators is suited to identifying which dimensions of trust matter and why, not to establishing how large an accuracy improvement, or what specific interface change, would be sufficient to shift adoption decisions.

A Case-Study Lens on Trust Evaluation Methodologically, this study treated each educator as a bounded case rather than as a data point in an aggregate usability score, which is what allowed the orthogonal-trust pattern to surface at all: a purely quantitative pre/post design comparing an aggregate “trust” score across conditions would have masked the fact that privacy and accuracy were moving independently of one another within the same participants. Triangulating think-aloud interaction, observer fieldnotes, and structured Likert items let us cross-check each construct against multiple independent evidence sources rather than relying on self-report alone (Table 3); the low internal consistency we observed for several constructs (e.g., $\alpha = 0.47$ for Privacy and Data Comfort, driven by the instructor-material/student-data split in §6.5) itself became an analytic finding rather than a nuisance to be explained away, since it pointed to a real, substantively meaningful split in how educators reasoned about different categories of data. For learning-analytics research evaluating human-facing AI systems with small, purposively

sampled stakeholder groups, we suggest this case-study orientation – treating internal heterogeneity as data rather than noise – as a transferable methodological stance, independent of the specific tool evaluated here.

Limitations Our findings reflect an initial qualitative probe with seven computing educators ($N = 7$), intended to surface emergent trust dimensions rather than estimate population parameters ($\alpha = 0.84$ for Accuracy; $\alpha = 0.79$ for Trust). The study evaluated a single 1.5B Qwen model running via Apple MLX; results across alternative model architectures, parameter scales (e.g., 3B–7B), and non-Apple hardware may differ. Finally, our data capture educator perceptions and immediate interaction dynamics rather than longitudinal student learning outcomes.

8. Conclusion

We developed TOOL, a locally deployed small language model (SLM) system that executes entirely on-device without internet connectivity and adapts to local course materials via no-code LoRA fine-tuning. We evaluated the system through automated pedagogical benchmarks and an in-depth qualitative case study with seven computing educators. On the automated benchmark, the fine-tuned TOOL model was 12.4 times as likely to provide stepwise instructional guidance as the base model under the content-focused rubric, while its aggregate MMLU accuracy was 0.2 percentage points lower. The similar aggregate accuracy suggests that fine-tuning did not materially reduce performance on the evaluated MMLU questions.

In the educator evaluation, local-first data privacy was the system’s most decisive operational asset: all seven participants felt comfortable uploading course documents (Q27, $M = 4.29$), and six of seven participants reported that on-device processing increased their willingness to integrate the tool (Q28, $M = 4.43$). Conversely, factual accuracy and parametric leakage emerged as the primary operational bottleneck: all seven participants disagreed that the model avoided generating information beyond their uploaded materials (Q6, $M = 1.86$), with every participant encountering hallucinations during testing. Composite trust sat below the scale midpoint ($M = 2.77$), with six of seven participants unwilling to deploy or use model outputs without independent verification. While participants were uniformly neutral on whether the assistant respected pedagogical intent (Q24, $M = 3.00$, $SD = 0.00$), direct observations suggested that the model often avoided direct answer-dumping but struggled to sustain curriculum-grounded, multi-turn guidance. Traceable citation infrastructure was identified by five of seven educators as the critical prerequisite for operational usability.

These findings demonstrate the empirical reality of orthogonal trust in educational AI: on-device SLMs can dismantle the data-sovereignty and surveillance barriers that restrict cloud-based models in educational institutions, but privacy guarantees operate independently from epistemic trust. Resolving data protection concerns does not compensate for factual unreliability. Until compact language models can reliably constrain their parametric leakage and signal factual uncertainty, the accuracy ceiling of local edge models remains the binding constraint on classroom adoption.

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Declaration of Conflicting Interest

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Supplementary Materials

Full survey instrument, protocols, codebook, and technical specifications (from Section 3) are available at:

[Orthogonal-Trust-The-Independence-of-Privacy-and-Accuracy-in-Instructor-Evaluations-of-AI-Tools-B674](#)

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